

Project Close-out Report (PCR)

Open Source Integration of Hummingbot with Cardano for Enhanced DeFi Liquidity

31 August 2025

Project Information

Project Name: Open Source Integration of Hummingbot with Cardano for Enhanced DeFi
IdeaScale URL: Catalyst page
Project Number: 1100212
Project Manager: Shay Gammer
Start Date: 15 March 2024
Completion Date: 31 August 2025

Challenge KPIs – Ecosystem Impact

- Improved liquidity across Cardano DEXs by enabling open-source algorithmic market-making.
- Increased developer adoption by providing standardized Cardano connectors within Hummingbot.
- Reduced barriers to algorithmic trading on Cardano by offering pre-built, tested connectors through a unified high-level API.

Project KPIs – Deliverables & Performance

- Gateway design & specification: published architecture and specification for Cardano–Hummingbot gateway middleware.
- Two Spot DEX connectors: delivered Cardano connectors for Minswap AMM DEX and SundaeSwap DEX.
- Testing & refinement: conducted unit and integration tests, ensuring execution parity with reference UIs.
- Documentation & integration: submitted connectors to the main Hummingbot codebase, with user documentation and configuration guides.
- Gateway repository: open-sourced at AIQUANT-Tech/HummingbotGateway.

Key Achievements (Collaboration & Impact)

- Two production-ready connectors: traders can now run Hummingbot strategies directly against Minswap and SundaeSwap.
- Standardised gateway architecture: added necessary functionality to support transactions under Cardano's eUTXO model. Middleware enables additional DEX connectors with minimal friction.
- Community adoption: early testers validated order execution and arbitrage strategies across multiple Cardano DEXs.
- Liquidity enhancement: algorithmic market-makers can deploy cross-exchange strategies, narrowing spreads and increasing trading depth.
- Collaboration with Hummingbot Foundation: ensured connectors meet community standards and merged into the global codebase.

Key Learnings

- eUTXO model adds complexity: handling multiple inputs/outputs in swaps required deeper gateway logic than account-based chains.
- DEX router diversity: AMM vs CLMM differences demanded modular connector design.
- Testing on-chain strategies is resource-intensive: simulation environments and pre-prod testnets were essential.
- Community contribution accelerates progress: engaging Hummingbot open-source developers improved connector quality and reduced maintenance overhead.
- Liquidity bootstrapping needs incentives: connectors alone are insufficient; DEXs may need reward programs to attract operators.

Next Steps

- Mainnet adoption push: work with Cardano DEX teams to promote connector availability to liquidity providers.
- More connectors: extend gateway support to WingRiders and MuesliSwap.
- Strategy templates: publish open-source configs tailored for Cardano (e.g., cross-DEX arbitrage, order-book replication).
- DAO collaboration: explore alignment between Cardano DeFi DAOs and Hummingbot governance for liquidity programs.
- Long-term maintenance: transition connector stewardship to the Hummingbot community with Cardano ecosystem backing.

Final Thoughts

This project facilitates open-source algorithmic trading on Cardano. By integrating Cardano DEXs into the global Hummingbot ecosystem, traders and liquidity providers gain the same tools available on other blockchains, making Cardano DeFi more competitive and accessible.

References & Links

- GitHub repo: github.com/AIQUANT-Tech/gateway
- Documentation:
 - SundaeSwap User Guide
 - Minswap User Guide
- Close-out video: [YouTube](#)